DFR@RMA Call 2026

Department:

SIGMA
Signal **I**nterference – **G**NSS-network **M**onitoring and **A**nalysis

Domain	☐ ***DAP*** ~~☐ ***MSP*** ☐ ***HFM*** ☐ ***DAP/MSP***~~
Focus Area	A1: Space Technologies & A3: Sensor Technologies
Project Acronym	SIGMA
Promotor	*Alain MULS*
Known researcher(s)	/
Number of researcher(s) to be hired ***Additional details should be included in table A, section 4.3 (e.g. level, pay scale, etc.).***	*1 researcher*
Total amount of Person Months requested	48 PM
FPP Security level	☐ *Unclass* ~~☐ *Confidential* ☐ *Secret NATO* ☐*Secret EU*~~
Project Security level	☐ *Unclass* ~~☐ *Confidential* ☐ *Secret NATO* ☐*Secret EU*~~
Duration of the project	48 Months
Requested start date of the project	*September / October 2026*
Total budget requested	*The total amount will be calculated by RSWO based on the data in the table in section 4.3.*

Summary of the project in laymen's terms *[max 5 lines]*
Protecting the L-band frequencies used by GNSS is vital for national security as it underpins critical infrastructure ensuring societal resilience against disruptions. Militarily, these frequencies enable precise navigation, targeting, and troop coordination, providing a significant strategic advantage. Interference or jamming of these signals can cripple both civilian and defense operations, leading to economic losses and jeopardizing national defense capabilities. Investment in protection ensures the reliability and integrity of these essential services against both intentional and unintentional threats. Ultimately, safeguarding the L-band strengthens a nation's overall security posture and technological sovereignty.

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1 PRESENTATION OF THE APPLICANT

1.1 Research activities

The research unit GENA (Geodesy & Navigation) focuses on Global Navigation Satellite Systems (GNSS) investigating the accuracy and reliability of different navigation services, interference resistance and spoofing detection of navigation signals, the benefits of multi-constellation and multi-frequency positioning and new signals, such as Galileo's Public Regulated Service (PRS). Next to real-time kinematic (RTK) and precise point positioning (PPP) techniques, the RU GENA is also looking into new low earth orbiting (LEO) navigation constellations as an alternative complementary positioning system for GNSS.

The National Geographic Institute (NGI) is the national mapping agency responsible for producing and updating topographic and geographic reference data for Belgium. The NGI is responsible for the national realization of the European ETRS89 coordinate system in Belgium. With this expertise, NGI plays a crucial role in monitoring and controlling the coordinate integrity of regional RTK GNSS networks (FLEPOS, WALCORS & GPSBRU). In essence, the NGI ensures the quality, accessibility, and interoperability of the high-precision GNSS data provided by the regional systems, making it a valuable resource for various applications requiring accurate location information across Belgium.

1.2 Experience in the research field

- ***3PfD - PRS Pilot Project for Demonstration***, European GNSS Agency (GSA), Jan 2018-Jun 2022.
  European security forces have an increased need for a reliable GNSS based Position, Navigation & Timing (PNT) services with authenticated and controlled access guaranteeing their PNT-superiority and protection from misuse activities (such as jamming) during operations. While Galileo PRS is not yet operational, it is essential to lay the foundation for it already today to build the confidence of the security forces and other potential user communities (UC) in the PRS navigation service and to ensure the emergence of a new European market dedicated to PRS services. Therefore, the main objective of this proposal is to demonstrate the ability of the PRS navigation service to guarantee continuous provision of PNT information to authorised governmental users during intentional or unintentional denial of open service (OS) navigation signals that are subjected to radio frequency interferences (RFI) vulnerabilities.

  Role of RMA: project coordinator
  Consortium formed by research institutes and industries from BEL, FIN, SWE, POL and GER.

- ***PETRUS - PRS European Tests for Robustness of User Segment***, European GNSS Agency (GSA), Jun 2018-May 2022.
  The overall objectives of the activities defined by the consortium answer the needs expressed by the GSA in the call for grant:
  - To monitor and evaluate the Galileo PRS quality of service (navigation performance, security performance, robustness), along with the progressive deployment of the Galileo system, assessing access control and all related operational PRS procedures, analysed with respect to the available functionalities and robustness, throughout laboratory and on the fields tests in user-oriented scenarios
  - To contribute to the consolidation of the Common Minimum Standards (CMS) implementation
  - To compare PRS applications to other Galileo services applications
  - To share with the program and among all Competent PRS Authorities (CPA) the results and the return of experience gathered through the consortium's activities
  - To benchmark PRS against other navigation/timing systems and propose a preliminary navigation policy
  - To initiate actions to promote the use of PRS services towards the PRS community and decisions makers

  Role of RMA: consortium member
  Consortium formed by research institutes and industries from FRA (project coordinator), BEL, SPA, ITA, POR.

- ***BE-GPIOS - Belgian Galileo PRS In Orbit validation Service***, BELSPO, Jan 2015-Jun 2016
  The Galileo PRS is an encrypted navigation service designed to be more resistant to jamming, involuntary interference and spoofing. It is similar to other Galileo services but ensures Continuity of Service (CoS) to authorised users when access to other navigation services is denied, increasing the likelihood of continuous availability of the Signal-in-Space (SiS) and provides an authenticated PNT service. The Binary Offset Code (BOC) modulation used by the PRS navigation signals moves the signal power away from the band centre, thus offering the potential for better code-tracking accuracy and multipath rejection. The high BOC modulation order of the PRS navigation signals, and the L1A signal in particular, are sensitive for false tracking of correlation peaks, phenomena observed during PRS by ESTEC. To obtain better insight in the environmental conditions and the correlation channel causing these false locks, the BE-GPIOS project proposes a methodology to benchmark the PRS signal tracking and PNT performance in operational environments.

  Role of RMA: perform the project under direction of European Space Agency (ESA)

1.3 Promotor profile and skills

The specific skills a project promoter must possess to carry out a scientific program are in-depth knowledge, technical proficiency, project management and organisation skills, data analysis and visualisation, good communication skills and strategic thinking and vision. In the specific context of geodesy and navigation using GNSS systems, the promotor would additionally benefit from GNSS expertise (understanding of GNSS principles, signal processing, error sources, and different GNSS constellations), geodetic principles (datum and coordinate systems), proficiency in GNSS data processing methodology and software and experience with GNSS reference networks.

In summary, the ideal promoter should possess a blend of strong scientific acumen, effective management skills and excellent communication abilities, all tailored to the specific demands of the research project.

Annexes to be included

- *Academic CV of the main promotor: I only have a PDF version of my CV, so I cannot include it here using the "insert object" method.*

2 PRESENTATION OF THE PROJECT

2.1 Context

Society and defence increasingly depend on PNT services[1] from GNSS systems, indispensable for modern defence, navigation and critical infrastructure. However, the vulnerability of navigation signals to interference —such as jamming and spoofing— poses significant risks to national security and operational continuity[2]. Addressing this challenge, this project proposes the development of methodologies for GNSS-network monitoring and analysis, and of a GNSS interference monitoring application leveraging GNSS Continuously Operating Reference Stations (CORS) of all Belgian real time kinematic GNSS (RTK-GNSS) networks (FLEPOS in Flanders, WALCORS in Wallonia and GPSBRU in Brussels capital region).

This project aims to establish a monitoring tool that utilizes GNSS data to detect, analyse and classify interference through navigation signals measurement (e.g. signal-to-noise ratio (SNR)) and observed station coordinate variations. The application will combine algorithms for anomaly detection with intuitive visualization features to support decision-making and threat mitigation. By providing warnings and actionable insights, the project enhances resilience in GNSS-reliant defence operations and civilian applications.

The project's technical focus includes integrating data from distributed GNSS networks, implementing robust interference diagnostics, and evaluating the tool through controlled experiments. Key outcomes include guidelines for deployment, recommendations for integrating monitoring capabilities into Belgium's national critical PNT infrastructure and providing key players, such as the Belgian Institute for Postal Services and Telecommunications (BIPT), the Competent PRS Authority (CPA) and Belgian Defence, support for setting up a Belgian wide territory GNSS monitoring tool. The project will also contribute to the EU GNSS Interference Task Force (EGITF) from the Space Programme Committee – Galileo EGNOS Configuration (SPC-GEC).

By monitoring GNSS interference, this project supports national defence and security priorities and contributes to technological leadership in critical geospatial intelligence. Aligned with the Belgian Royal Higher Institute for Defence's focus areas of Data Acquisition and Processing (DAP), the project addresses both technological innovation and strategic needs.

2.2 State of the art and innovative character

Introduction: The Criticality of GNSS and the Interference Threat

The increasing reliance on GNSS across a wide spectrum of applications, from personal navigation and transportation to critical infrastructure (communications, electricity,…) and national security, has brought the vulnerability of these systems to radio frequency interference (RFI) into sharp focus. Both unintentional interferences, stemming from various electronic devices, and intentional attacks, such as jamming and spoofing, pose significant threats to the integrity and reliability of GNSS signals. The increasing sophistication of these threats necessitates the development and implementation of robust and effective interference detection techniques.

A surge in GNSS signal interference is threatening civil aviation, maritime safety, and global telecom networks, prompting a joint warning from three UN agencies calling on governments to act[3]. The agencies are also urging the development of new systems and infrastructures that could reduce this threat. In this rare joint statement, the heads of the International Telecommunication Union (ITU), the International Civil Aviation Organization (ICAO), and the International Maritime Organization (IMO) said they were "gravely concerned" about the increasing number of incidents involving harmful interference with the Radio Navigation Satellite Service (RNSS), which includes global navigation satellite systems such as GPS. The three agencies urged member states to bolster the resilience of essential navigation and timing infrastructure that depends on satellite signals[4].

In the context of the Galileo Public Regulated Service (PRS), the Competent PRS Authority (CPA) is responsible for managing and monitoring the PRS service within a Member State's territory, including monitoring for and addressing electromagnetic interference[5]. Among other tasks the CPA is responsible for continuously monitoring compliance with technical requirements and for managing any detected harmful electromagnetic interference affecting the PRS on its national territory.

The Belgian Institute for Postal Services and Telecommunications (BIPT)[6] is the federal regulator responsible for the electronic communications market, the postal market, the electromagnetic radio frequency spectrum, as well as audio-visual media services and video platform services in the Brussels-Capital Region. One of BIPT's

missions is to distribute radio frequencies to ensure that they are used efficiently and to constantly monitor the radio waves to stop harmful interference.

The SIGMA proposal aims to develop advanced GNSS-network monitoring and analysis methodologies that enable Belgium to perform interference monitoring over its entire territory using the dense regional networks of the regional governments.

State of the Art in GNSS Interference Monitoring

Spectrum analysis visualises the signal powers across a frequency band allowing users to see the presence and strengths of interfering signals. Once interference is identified, appropriate mitigation techniques can be implemented, such as filtering and interference mitigation software[7], usually implemented by receiver firmware.

Eurocontrol uses Automatic Dependent Surveillance – Broadcast (ADS-B), a surveillance system used by commercial aircraft, for GNSS RFI Monitoring[8]. ADS-B broadcasts position and velocity messages based on positions from certified GNSS based position estimates. A RFI jamming source can disrupt aircraft operations and result in unavailability of GNSS service. This sudden loss of navigation can be monitored through ADS-B reports on the ground side. Therefore, by monitoring the ADS-B outputs on the ground side, we can identify instances when interference impacts an aircraft's positioning and navigation capabilities. This technique is however only available for commercial flights supervised by Eurocontrol.

In the signal processing domain[9], some of the proposed algorithms include the monitoring of the receiver power spectrum, of the automatic gain control level, or the signal correlation peak. These solutions all imply access and modification of the receiver firmware and thus increase the complexity of the detection method.

An interference detection system using GNSS receiver independent exchange (RINEX) data from the entire SWEPOS[10] RTK network effectively detects GNSS interference by examining the signal-to-noise ratio (SNR) of GNSS frequency bands. Comparing the SNR of tracked satellites against historical data, the system detects RFI by statistically characterizing the SNR of simultaneously tracked satellites. The system monitors for unexpected changes in the SNR values and characterizes how these changes correlate between satellites tracked at the same time. Furthermore, based on the SNR characteristics across satellites, the system distinguishes between RFI and non-RFI sources. A similar system is developed in Poland by a consortium of the National Institute of Telecommunications and the Head Office of Geodesy and Cartography, funded under ESA's NAVISP program[11]. However, these solutions are often limited to proprietary binary data streams.

Key Performance Indicators (KPI) within project SIGMA

Project SIGMA will base its interference detection on data collected by the regional GNSS networks available in the RINEX format. This format contains for each observed signal the pseudo-distance, integrated phase, Doppler frequency and SNR values. Being responsible for Belgians geographic reference, the NGI determines the exact coordinates of the regional reference stations. Therefore, the relevant performance metrics are carrier to noise ratio (C/N0), position error and consistency of the measurements as the most useful KPIs SIGMA can examine.

KPI 1. Signal-to-Noise Ratio (SNR)

SNR is a key performance indicator quantifying the strength of the received satellite signal relative to the background noise level within a given bandwidth. In the context of GNSS, SNR is often expressed as Carrier-to-Noise Density Ratio (C/N0) in units of dBHz, which represents the signal power to noise power ratio normalized to a 1 Hz bandwidth. The inherent weakness makes GNSS signals particularly vulnerable to interference from even relatively low-power interference sources.

Literature shows several effective methods for SNR-based interference detection. Lebrun et al.[12] demonstrated the effectiveness of using seasonal and trend decomposition using LOESS (STL) combined with autoregressive integrated moving average (ARIMA) models to detect anomalies in C/N0 time series. Thompson et al.[13] proposed using statistical analysis of C/N0 variations across multiple satellites to identify interference patterns. Amin et al.[14] showed that time-frequency analysis of C/N0 measurements can distinguish between different types of interference. These methods can be combined to create a robust detection system that leverages both temporal patterns and statistical analysis of signal quality metrics.

KPI 2. Continuously Operating Reference Station (CORS) coordinates

Under normal conditions, a CORS station's calculated position remains highly stable over time, with only minor variations. By continuously monitoring the calculated coordinates of a CORS station, any significant deviation from its known position can indicate a problem. RFI or jamming typically increases the noise and uncertainty in GNSS measurements, resulting in larger variations or increased scatter in the calculated

coordinates. While the calculated position might not be grossly wrong, the precision degrades. Spoofing, on the other hand, deliberately transmits false signals to mislead the receiver. This can cause the calculated CORS station coordinates to shift gradually or abruptly away from their true location. Because the position of a CORS station is fixed, any movement is a clear indicator of spoofing. Techniques like wavelet analysis or short-time Fourier transform can be used to examine the frequency content of the coordinate variations. This can help distinguish between different types of interference (e.g., narrowband vs. broadband jamming) and potentially identify the characteristics of a spoofing signal.

Recent research has advanced coordinate-based detection methods. Psiaki and Humphreys[15] developed techniques to detect spoofing by analysing position solution consistency and receiver clock behaviour. Lewis et al.[16] proposed a cross-layered integrity processing approach that combines position solutions with other GNSS observables. Abraha et al.[17] demonstrated the effectiveness of using coordinate time series analysis to detect both intentional and unintentional interference. These methods can be enhanced by incorporating machine learning techniques to improve detection accuracy and reduce false alarms as demonstrated in Van der Merwe et al.[18].

KPI 3. Network Analysis

The Belgian regional GNSS networks boasts one of the densest GNSS networks globally. Comparing received signals between neighbouring stations offers the possibility to detect an RFI event by comparing simultaneous SNR values between these stations.

If one CORS station shows a position shift while neighbouring stations remain stable strengthens the evidence that the affected station is experiencing interference or spoofing. Analysing the spatial pattern of coordinate variations across the network can also help distinguish between localized interference and a more widespread spoofing attack.

Network-based detection methods have been extensively studied. Qiao et al.[19] reviewed various network-based approaches, highlighting the effectiveness of spatial correlation analysis between stations. Van der Merwe et al.[xviii] demonstrated that low-cost commercial-off-the-shelf (COTS) receivers can be effectively used in a network configuration to detect and classify interference. Yinhui et al.[20] proposed a blanket interference monitoring approach that leverages the spatial distribution of monitoring stations to improve detection accuracy. These methods can be particularly effective in Belgium's dense network, where the close spacing of stations provides excellent spatial resolution for interference detection and localization.

KPI 4. Multi-GNSS and Multi-Frequency Analysis:

The redundancy of multiple GNSS constellations and multiple frequencies tracked by modern CORS stations can be used to enhance detection:

- Spoofing is often easier to implement on some signals than others. By comparing the consistency of position solutions derived from different constellations and frequencies, spoofing can be detected. For example, if GPS L1 is being spoofed, but Galileo E1 is not, the discrepancy will be apparent.
- Jamming may affect different frequencies and constellations differently. Analysing these differences can provide clues about the jammer's characteristics.

Multi-constellation and multi-frequency analysis has proven effective in interference detection. Psiaki and Humphreys[15] showed that comparing solutions from different constellations can reveal spoofing attempts. Lewis et al.[16] demonstrated that cross-layered integrity processing across multiple frequencies improves detection reliability. Recent work by Qiao et al.[19] and Van der Merwe et al.[19] has shown that combining data from multiple constellations (GPS, Galileo, GLONASS, BeiDou) significantly enhances the ability to detect and classify interference types. These methods can be particularly valuable in Belgium's network, where modern CORS stations track multiple constellations and frequencies.

Innovative character

The SIGMA project distinguishes itself through its integration of novel methodologies and the strategic use of existing GNSS infrastructures. Unlike conventional approaches focused on spectrum sensing or aviation-based monitoring, SIGMA leverages Belgium's high-density network of RTK GNSS CORS to detect and analyse signal interference using openly accessible GNSS observables, particularly SNR data and coordinate solution stability metrics. The originality of the project stems from several innovations:

- **Network-wide Monitoring:** Instead of isolated monitoring points, SIGMA implements a system-wide approach, correlating anomalies across multiple CORS stations in enhancing detection reliability and spatial resolution.

- **Multi-Constellation and Multi-Frequency Analysis:** By concurrently analyzing GPS, Galileo, GLONASS, and BeiDou signals across multiple frequency bands, SIGMA increases resilience against both narrowband and wideband interference, addressing spoofing and jamming threats with greater sensitivity.
- **Statistical and Predictive Modelling:** The project introduces advanced statistical baselining techniques, including time series anomaly detection based on seasonal and trend decomposition and classification, minimizing false positives and enabling detection of threats.
- **Infrastructure-Compatible Deployment:** SIGMA's detection methodologies are designed to operate without modifications to existing GNSS receivers, using standard RINEX data. This greatly reduces deployment cost and complexity, resulting in rapid scalability. It also allows easy densification of the sensing network with e.g. low-cost COTS receivers.
- **Actionable Outputs for National Security:** Beyond mere detection, SIGMA aims to integrate interference event reporting into operational decision-making, providing alerts, visual dashboards, and strategic recommendations for stakeholders including defense, aviation, emergency services, and telecommunications.

By combining these elements, SIGMA advances the state-of-the-art and establishes a sustainable, scalable, and operational framework for GNSS interference monitoring and national PNT security enhancement.

2.3 Project summary

The SIGMA project – "Signal Interference – GNSS-network Monitoring and Analysis" – addresses the increasing threats to PNT services caused by GNSS interference, including jamming and spoofing. Motivated by the reliance of modern defence, transportation, and critical infrastructure sectors on GNSS signals, the project aims at strengthening national resilience by detecting and analyzing GNSS signal anomalies.

The core innovation of SIGMA lies in leveraging Belgium's extensive RTK GNSS networks (FLEPOS, WALCORS, and GPSBRU) to monitor interference using openly accessible GNSS observables, particularly Signal-to-Noise Ratio (SNR) fluctuations and coordinate solution stability. Through advanced statistical modelling, cross-station anomaly detection, and multi-constellation multi-frequency analysis, SIGMA can deliver a low-cost, scalable, and operationally ready solution without the need for hardware modifications.

The expected results include an interference detection platform and a visual analytics dashboard paving the way forward to an automated reporting API. For the defence sector, the project enhances situational awareness of GNSS vulnerabilities, support rapid incident response, and contribute to the protection of critical PNT-dependent operations within the extent of the GNSS monitoring network. Additionally, SIGMA aligns with broader European initiatives, such as the EU GNSS Interference Task Force (EGITF), reinforcing Belgium's strategic role within the EU's security architecture.

The project will be carried out in three phases: (1) defining technical requirements and pre-processing of GNSS data for analysis; (2) development and validation of anomaly detection and classification algorithms; and (3) integrating the framework into a visualization prototype which can be extended with alerting capabilities. Validation will be based on data from the Belgian RTK GNSS networks, with scalability toward national deployment.

The framework can serve several stakeholders in the Belgian defense and security sector, such as BIPT, Crisiscentrum, CPA (NVO/ANS).

2.4 Project objectives

Objective	Specific	Measurable	Achievable	Realistic	Time-bound
O1: Build a GNSS data collection and processing pipeline	Establish a processing system from RINEX-data to a ready-to-use data format.	Functional pipeline tested with at least 95% data availability over a month.	Utilizes existing GNSS network access agreements.	Compatible with current infrastructure.	Complete within the first 15 months.
O2: Develop a signal-based interference detection algorithm	Design algorithms to detect interference.	Anomaly-detection on simulated data with false positive/negative rates	Leverages statistical models validated in similar international networks.	Feasible given available datasets and open-source tools.	Algorithms ready by month 24
O3: Develop a coordinate-based anomaly detection algorithm	Design algorithms to detect interference.	Anomaly-detection on simulated data with false positive/negative rates	Leverages statistical models validated in similar international networks.	Feasible given available datasets and open-source tools.	Algorithms ready by month 28
O4: Implement a multi-station network correlation module	Compare anomalies across stations to validate events spatially.	Anomaly-detection on simulated data with false positive/negative rates	Builds on standard statistical correlation techniques.	Realistic considering network density.	Module functional by month 38
O5: Classify detected anomalies	Interference type classification.	Anomaly-detection on simulated data with false positive/negative rates	Builds on standard statistical classification techniques.	Feasible given available datasets and open-source tools.	Classification possible by month 40
O6: Develop a visualization dashboard and API for alerts	Create a web-based monitoring platform and automated alert API.	Dashboard operational and API generating alerts.	Uses existing technologies (e.g., Grafana, REST APIs).	Feasible with project team expertise.	First prototype by month 40; final version by month 48.
O7: Validate system performance and define operational guidelines	Test the system on historic data; draft a recommendation report for Defense users.	Successful detection and classification of 95% of known disturbances during final validation.	Builds on operational experience and stakeholder feedback.	Feasible with project team expertise.	Final validation and report delivery by month 48.

3 RELEVANCE AND POTENTIAL IMPACT ON DEFENCE

3.1 Potential impact of the proposal in light of the expected outcomes

The SIGMA project addresses critical challenges in protecting and monitoring GNSS signals that are fundamental to both military operations and civilian critical infrastructure. The project's impact spans multiple dimensions that directly align with Defence's focus areas in Space Technologies (A1) and Sensor Technologies (A3):

Strategic Impact:

1. Development of a nationwide GNSS interference monitoring capability that enhances Belgium's technological sovereignty and strategic autonomy in protecting critical PNT infrastructure
2. Strengthening of Belgium's contribution to European GNSS security initiatives, such as ESA NAVISP programme
3. Support for the Belgian Competent PRS Authority (CPA) in fulfilling its mandate to monitor and protect Galileo PRS services across Belgian territory

Operational Impact

1. Detection and characterization of GNSS interference events across Belgium using existing CORS infrastructure (FLEPOS, WALCORS, GPSBRU)
2. Enhanced situational awareness of GNSS vulnerabilities for military operations, enabling proactive measures against potential interference threats
3. Development of standardized metrics and methodologies for assessing GNSS signal quality and interference patterns
4. The developed SIGMA application grants user-friendly access to these features

Technical Innovation:

1. Development of advanced SNR-based interference detection algorithms using seasonal and trend decomposition (STL) combined with ARIMA models for anomaly detection in C/N0 time series
2. Implementation of sophisticated coordinate stability monitoring techniques using wavelet analysis and short-time Fourier transform (STFT) for detecting and characterizing different types of interference
3. Creation of novel network analysis methodologies leveraging Belgium's dense GNSS reference station infrastructure for spatial correlation and interference source localization
4. Integration of multi-constellation (GPS, Galileo, GLONASS, BeiDou) and multi-frequency monitoring capabilities with statistical analysis of signal variations across satellites

Knowledge Development

1. Building expertise in GNSS interference monitoring and analysis
2. Creation of a knowledge base for GNSS threat assessment and mitigation strategies
3. Development of the SIGMA application

The project's deliverables will provide Defence and critical infrastructures with concrete tools and capabilities for protecting critical GNSS-dependent operations while contributing to national and European efforts in securing space-based PNT services.

3.2 Plans to maximise the impact of the project

The SIGMA project will implement a comprehensive valorisation strategy across multiple domains:

Academic & Defence Integration:

1. Publication of findings in high-impact journals and presentations at key conferences (ION GNSS+, European Navigation Conference)
2. Integration of project outcomes into RMA courses, particularly those related to GNSS technology and security
3. GNSS interference monitoring application for academic and Defence use
4. Regular briefings to RHID

Stakeholder Engagement:

1. Strong collaboration with the NGI
2. Regular briefings with the CPA
3. Establish contact lines with FLEPOS, WALCORS, GPSBRU, critical infrastructure operators such as ELIA, telecommunications companies, BIPT, Space Security Centre, ACOS transformation

Technology Transfer & Knowledge Preservation:

1. Development of a modular software framework for GNSS interference monitoring
2. Creation of user-friendly visualization tools and dashboards
3. Documentation of API's and interfaces for system integration
4. Maintenance of a knowledge base on GNSS interference patterns and detection methods
5. Developments will be shared publicly through git repositories

3.3 Contribution to Defence's R&T operational objectives

High-Quality Research:

1. Development of novel algorithms for GNSS interference detection and analysis
2. Creation of advanced monitoring methodologies using Belgium's dense GNSS network
3. Implementation of multi-constellation and multi-frequency monitoring capabilities
4. Integration of machine learning techniques for interference pattern recognition

Strengthening Cooperation:

1. Partnership with CPA and BIPT on spectrum monitoring, engagement with GNSS network operators
2. Partnership with National Geographic Institute of Belgium,
3. Technical exchanges with international partners on GNSS security (e.g. ESA NAVISP, EUSPA, ...)

Improving Communication:

1. Regular briefings to RHID on project progress
2. Active participation in academic and technical conferences

Valorisation:

1. Publication of research findings in high-impact journals
2. Development of operational tools and monitoring frameworks
3. Integration of project outcomes into RMA courses
4. Creation of a knowledge base for GNSS threat assessment
5. Development of a modular software framework for GNSS interference monitoring (open source)

4 COHERENCE BETWEEN RESEARCH OBJECTIVES AND METHODOLOGY

4.1 Work plan

4.1.1 Link between work packages

WPX: Project Coordination and Management

- **WP Leader:** RMA (Alain MULS)
- **Tasks**:
  a. Project planning and coordination (Months 1-48)
     i. Project timeline and milestone tracking
  b. Communication (Months 1-48)
     i. Yearly reports to KHID
     ii. Quarterly meetings RMA-NGI
     iii. Yearly meetings with all participants
- **Participants**: RMA, NGI, (BIPT, CPA, ACOS Transformation)
- **Effort:** 48 PM
- **Deliverables:** Project management plan, progress reports, meeting minutes
- **Follow-up indicators:** Work packages completion
- **Risk Assessment:** None

WP1: Literature Study and Requirements Analysis

- **WP Leader**: RMA
- **Tasks:**
  a. Literature review on GNSS interference detection (Months 1-6)
     i. Review of STL and ARIMA-based C/N0 analysis methods
     ii. Analysis of wavelet and STFT techniques for coordinate stability monitoring
     iii. Study of network-based detection approaches and spatial correlation methods
  b. Analysis of existing monitoring systems (Months 1-6)
     i. Review of national and international GNSS monitoring initiatives
     ii. Analysis of multi-constellation and multi-frequency monitoring approaches
     iii. Study of signal quality metrics and interference classification methods
  c. Technical requirements definition (Months 3-9)
     i. Specification of RINEX processing requirements
     ii. Definition of signal quality metrics and thresholds
     iii. Requirements for multi-constellation support (GPS, Galileo, GLONASS, BeiDou)
- **Participants:** RMA, NGI
- **Effort:** 9 PM
- **Deliverables:** Literature review report, system requirements specification
- **Follow-up indicators:** Completeness of literature review, technical requirements validation
- **Risk Assessment:**
  o Risk: literature not available
    - Impact: Medium
    - Probability: Zero to Low
    - literature concerning this topic is readily available on well-known platforms

WP2: Data Collection and Processing Pipeline

- **WP Leader:** RMA
- **Tasks**:
  a. Data Collection and Processing Pipeline (Months 3-6)
     i. Data wrangling
     ii. Data validation and quality checks
  b. Getting acquainted with regional networks (Months 6-9)
     i. Secure data transfer from CORS station to NGI servers
     ii. Network health monitoring
  c. Data pre-processing and quality control (Months 9-15)
     i. Statistical outlier detection
     ii. Signal quality metrics development
- **Participants**: RMA, NGI, (FLEPOS, WALCORS, GPSBRU)
- **Effort:** 15 PM
- **Deliverables:** Functional data pipeline, quality control reports
- **Follow-up indicators:** Data availability (>95%), processing time, error rates
- **Risk Assessment:**
  - Risk: Data access restrictions
    - Impact: High
    - Probability: Zero to Low
    - NGI has the resources to provide the data
  - Risk: Data quality degradation
    - Impact: High
    - Probability: Zero to Low
      - The national GNSS CORS are regularly checked and well-maintained

WP3: Signal-Based Interference Detection Development

- **WP Leader:** RMA
- **Tasks**:
  a. Signal strength analysis (Months 10-24)
     i. Implementation of Seasonal and Trend decomposition for C/N0 time series analysis
     ii. ARIMA modelling for anomaly detection in signal quality metrics
     iii. Multi-frequency band analysis with statistical characterization
  b. Multi-station C/N0 correlation analysis (Months 24-34)
     i. Spatial correlation algorithms for network-wide detection
     ii. Implementation of cross-station validation methods
     iii. Development of interference source localization techniques
- **Participants**: RMA, NGI
- **Effort:** 24 PM
- **Deliverables:** Detection algorithms, validation reports, performance metrics
- **Follow-up indicators:** Detection accuracy, false positive rates, processing speed
- **Risk Assessment:**
  - Risk: Algorithm performance degradation
    - Impact: High
    - Probability: Medium
    - Mitigation: Extensive testing, performance benchmarks, adaptive algorithm tuning
  - Risk: High false positive/negative rates
    - Impact: High
    - Probability: Medium
    - Mitigation: Dynamic threshold adjustments, multi-stage validation, continuous performance monitoring

WP4: Position-Based Interference Detection Development

- **WP Leader:** RMA
- **Tasks**:
  a. Position stability monitoring (Months 20-28)
     i. Wavelet analysis implementation for coordinate variation detection
     ii. Short-Time Fourier Transform (STFT) processing for frequency domain analysis of position solutions
     iii. Development of position stability indicators and thresholds
  b. Multi-station correlation analysis (Months 28-38)
     i. Spatial correlation algorithms for network-wide detection
     ii. Implementation of cross-station validation methods
     iii. Development of interference source localization techniques
- **Participants**: RMA, NGI
- **Effort:** 24 PM
- **Deliverables:** Detection algorithms, validation reports, performance metrics
- **Follow-up indicators:** Detection accuracy, false positive rates, processing speed
- **Risk Assessment:**
  - Risk: Algorithm performance degradation
    - Impact: High
    - Probability: Medium
    - Mitigation: Extensive testing, performance benchmarks, adaptive algorithm tuning
  - Risk: High false positive/negative rates
    - Impact: High
    - Probability: Medium
    - Mitigation: Dynamic threshold adjustments, multi-stage validation, continuous performance monitoring

WP5: Interference Classification and Analysis

- **WP Leader:** RMA
- **Tasks**:
  a. Interference type classification (Months 34-40)
     i. Temporal: permanent, temporary, periodic
     ii. Spatial: national, regional, local
     iii. Type: intentional, unintentional
- **Participants**: RMA, NGI
- **Effort:** 6 PM
- **Deliverables:** Classification algorithms, pattern database, validation reports
- **Follow-up indicators:** Classification accuracy, false positive rates, processing speed
- **Risk Assessment:**
  - Risk: Classification accuracy degradation
    - Impact: High
    - Probability: Medium
    - Mitigation: Hybrid classification approach, expert validation

WP6: SIGMA Application

- **WP Leader:** RMA
- **Tasks:**
  a. SIGMA Dashboard development (Months 36-48)
    i. Monitoring interface
    ii. Interactive maps and signal displays
    iii. Event visualization and reporting
- **Participants**: RMA, NGI, BIPT
- **Effort**: 12 PM
- **Deliverables**: Operational prototype
- **Follow-up indicators**: System stability, response time
- **Risk Assessment**:
  - Risk: Integration failures
    - Impact: High
    - Probability: Medium
    - Mitigation: rollback procedures
  - Risk: User interface usability issues
    - Impact: Medium
    - Probability: Medium
    - Mitigation: iterative interface improvements

WPZ: Valorisation, dissemination, exploitation of results

- **WP Leader**: RMA
- **Tasks:**
  a. Academic publications (Months 12-48)
    i. Journal articles in navigation and aerospace
    ii. Conference papers and presentations
    iii. Technical reports and evaluations
  b. Conference presentations (Months 12-48)
    i. Technical presentations
    ii. System demonstrations
    iii. Training workshops
  c. Knowledge transfer (Months 24-48)
    i. Technical documentation
    ii. Git repository
    iii. SIGMA application
- **Participants**: RMA, NGI, (BIPT, CPA, ACOS Transformation)
- **Effort**: 36 PM
- **Deliverables**: Publications, presentations, software application
- **Follow-up indicators**: Publication impact, application usage/reviews
- **Risk Assessment**:
  - Risk: Publication delays
    - Impact: Low
    - Probability: Medium
    - Mitigation: Early publication planning, regular progress reviews, multiple publication venues
  - Risk: SIGMA application development
    - Impact: Low
    - Probability: Medium
    - Mitigation: Early application development if possible

4.1.2 Work planning and schedule: GANTT chart

Provide the planning in the form of a GANTT chart (start and finish dates for each WP). Please use the GANTT chart template provided.

4.1.3 Workload intensity in relation to work packages: GANTT chart

Complete the GANTT chart in accordance with the detailed work plan, tasks and deliverables described above.

- *Work intensity of each partner for each task (expressed in person-months [PM]), if more than one partner is involved in the project.*
- *Include for each partner the person-months funded by the project and the person-months funded by other sources (see notes).*

Remarks:

- *Partners include funded, non-funded partners and subcontractors.*
- *1 Person-month [PM] = 1 full-time equivalent [FTE] or 2 half-time equivalents over 1 month.*
- *Other sources of funding may include payment of salaries by institutions other than Defence and/or via other projects, voluntary contributions, etc. If a given task requires 7 person-months, and 6 months will be funded by the project, the 7th month must be included under 'other sources of funding'.*

4.2 Experts

1. Dr Carine BRUYNINX, Royal Observatory of Belgium, Head of the Central Bureau of the EUREF Permanent GNSS Network, carne.bruyninx@oma.be, +32 2 3730292.
2. Dr Tom WILLEMS, Senior Consultant CGI, tom.willems@ganda-sys.net, +32 496 127882.
3. Dr ir Frank BOON, R&D Group Manager at Septentrio, f.boon@septentrio.com., +32 16 300800.

RSWO is responsible for selecting the most appropriate experts to evaluate research projects. However, in specific cases, RSWO may consider experts proposed by the project promotors. Therefore, we kindly request that you suggest two scientific experts in the relevant field who could assess this proposal impartially and provide a rationale for your choice. Additionally, please include their contact details. It is essential that there is no conflict of interest between the promotor/researcher(s) and the proposed experts. Additionally, the proposed experts must be authorized to evaluate the project according to the security level mentioned on page 1.

Promotors also have the option to request the exclusion of up to two experts from the evaluation of their proposal. Should you wish to exercise this option, please provide the names of the experts along with their affiliated institutions.

4.3 Budget

The budget will exclusively cover the costs of R&D activities related to the project.

Please complete the tables below with information regarding the researcher(s), planned missions, operational costs, and required computer equipment. This will enable the Bur ProjInt to calculate the total budget for the project proposal as accurately as possible.

- *Further explanations for the categories listed in the table A below:*
  - *Researchers: only researchers dedicated to the implementation of the project (personnel recruited by Defence or Patrimony). If a researcher (including promotors, repetitors and military staff) is funded by another source, mention it in the GANTT chart.*
  - *Open contract computers: choose between two types of laptops and one type of desktop (technical specifications available in the file "DFR2026_Technical_characteristics_open_contract_computers" on the intranet and Google Drive). If none of these meet your needs, mention it in the list of equipment below.*
  - *Missions: mission-related expenses. A fixed amount of €3,000/year has been set for missions. If this amount is not sufficient, please explain the reason and the new amount requested.*
  - *Functioning costs: includes all purchases less than 10,000€ VAT included. This covers consumables, services, software, CIS and non-CIS equipment, meeting related to the project organization costs. (e.g. meals, drinks and external meeting room hire), dissemination of project results, and publication costs. Doctoral registration fees are eligible. Doctoral defence costs are eligible only if the defence takes place at the RMA. A fixed amount of €3,000/year has been set for operating costs. If this amount is not sufficient, please explain the reason and the new amount requested in table A. Then list in more detail the equipment required in table B.*

For all major equipment purchases (≥10,000€ VAT included), please use the 'DFR2026_investments_KIM' template. This list should cover all purchases for the entire duration of the project. Please make sure all columns in the Excel sheet are filled out correctly to ensure the document can be easily and efficiently used by the Matman.
For more information regarding the investments/KIM, please consult and use the explanation tab of the Excel file "DFR2026_investments_KIM".

Table A:	**2026**	**2027**	**2028**	**2029**	**2030**
Researchers					
Number	1	1	1	1	1
Level (Level D, C, Bachelor Niv. B, Master Niv. A, ir. or PhD)	*Niv. A ir.*	*Niv. A ir.*	*Niv. A ir.*	*Niv. A ir.*	*Niv. A ir.*
Pay scale (DC1, C1, B1, NA11, NA21)	*NA21*	*NA21*	*NA21*	*NA21*	*NA21*
Allocation in %	*100%*	*100%*	*100%*	*100%*	*100%*
Open contract computers:					
Laptop A					
Laptop B	*1*				
Desktop			1		
Accessories	*Headset, keyboard, USB-C hub (ethernet, USB-A and usb-C ports, DisplayPort and VGA)*		Monitor, External SSD >= 1 TB		
Missions					
Travels > 3000€ VAT included	Y	Y	Y	Y	Y
If Yes: justification	*Attending international conferences and workshops. Conferences outside Europe exceed 3keuro*	*Attending international conferences and workshops. Conferences outside Europe exceed 3keuro*	*Attending international conferences and workshops. Conferences outside Europe exceed 3keuro*	*Attending international conferences and workshops. Conferences outside Europe exceed 3keuro*	*Attending international conferences and workshops. Conferences outside Europe exceed 3keuro*
Functioning costs					
Functioning costs >	N	N	N	N	N

3000€ VAT included				
If Yes: justification	*Short explanation + expected amount*			

Table B: *List of required equipment (ONLY* <10,000 € VAT included). *For purchases exceeding €10,000, please complete the "DFR2026_investments_KIM" template.*

Description of the need:	*Estimated amount (if known):*
1) 4K monitor with integrated USB-C hub	*1 keuro*
2) Yearly subscription of software tools concerning data science, application development and AI (machine learning or LLM support)	*1 keuro/year*
3)	
4)	
5)	
6)	
7)	
8)	

5 ACRONYMS

ADS-B	Automatic Dependent Surveillance – Broadcast
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
BIPT	Belgian Institute for Postal Services and Telecommunications
BOC	Binary Offset Code
C/N0	Carrier-to-Noise Ratio
CMS	Common Minimum Standards
CORS	Continuously Operating Reference Stations
CoS	Continuity of Service
COTS	commercial-off-the-shelf
CPA	Competent PRS Authority
DAP	Data Acquisition and Processing
EGITF	EU GNSS Interference Task Force
ESA	European Space Agency
ESTEC	European Space Research and Technology Centre
ETRS89	European Terrestrial Reference System 1989
EUSPA	EU Agency for the Space Programme
FLEPOS	Flemish Positioning Services
GENA	Research unit Geodesy & Navigation
GNSS	Global Navigation Satellite Systems
GPS	Global Positioning System
GPSBru	GPS Brussels
ICAO	International Civil Aviation Organization
IMO	International Maritime Organization
ITU	International Telecommunication Union
KPI	Key Performance Indicators
LEO	Low Earth Orbiting
LOESS	Locally Estimated Scatterplot Smoothing
NAVISP	Navigation Innovation and Support Programme
NGI	National Geographic Institute Belgium
NVO-ANS	Nationale Veiligheidsoverheid – Autorité Nationale de Sécurité
OS	Open Service
PNT	Position, Navigation & Timing
PPP	Precise Point Positioning
PRS	Public Regulated Service
RFI	Radio Frequency Interferences
RHID	Royal Higher Institute for Defence
RINEX	Receiver Independent Exchange
RMA	Royal Military Academy
RNSS	Radio Navigation Satellite Service
RTK	real-time kinematic

SIGMA	Signal Interference – GNSS-network Monitoring and Analysis
SiS	Signal-in-Space
SNR	Signal-to-Noise Ratio
SPS-GEC	Space Programme Committee – Galileo EGNOS Configuration
STFT	short-time Fourier transform
STL	Seasonal-Trend decomposition using LOESS
SWEPOS	Swedish Positioning System
UC	User Communities
WALCORS	Wallonia Continuously Operating Reference System

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